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MelodyCare

An Automated Two-Stage Generative Framework for Converting Vietnamese
Commercial Music into Clinically-Bounded Therapeutic Audio for
Individuals with Autism Spectrum Disorder

Final Report

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August 2026

Abstract

Individuals with Autism Spectrum Disorder (ASD) frequently experience auditory hyperacusis, in which ordinary commercial music — with its unpredictable dynamics, vocals, harsh transients, and fast tempo — can trigger sensory overload rather than provide comfort. Existing therapeutic-music solutions either rely on slow, manual re-engineering by trained audio professionals (Neurologic Music Therapy editing), which does not scale to a specific song a specific individual wants to listen to, or on unconditioned generative audio models, which can produce calming-sounding audio but cannot preserve the identity of a requested song and offer no mechanism to guarantee hard clinical acoustic-safety bounds. This project, MelodyCare, addresses both gaps with a two-stage pipeline: a deterministic DSP stage (Demucs source separation, BPM-locked time-stretching, Butterworth spectral shaping, dynamic-range compression, legato smoothing, and low-frequency breathing modulation) that both builds paired training data and acts as a final deterministic safety net; and a conditioned generative stage that fine-tunes stabilityai/stable-audio-open-1.0 with two newly-built cross-attention conditioning modules — a CLAP-derived style-vector projection and a novel input-audio projection that lets the model condition on a full source track despite the pretrained checkpoint having no built-in audio-to-audio conditioning pathway. The model was fine-tuned via multi-GPU DDP on 775 paired 30-second chunks drawn from 78 Vietnamese commercial songs, using differential learning rates and zero-initialization on both new modules to prevent catastrophic forgetting. Evaluated against seven objective clinical-safety checks on 10 in-sample and 10 genuinely unseen held-out songs, the Stage 3 deterministic safety net closes five of seven checks at 100% on both populations and reliably corrects dynamic range from 0–1/10 pass at the generative stage to 10/10 after correction — direct evidence the safety-net design works as intended — while legato-overlap correction remains an open limitation, and an ablation study surfaces a genuine negative finding that the source-audio conditioning module's contribution could not be detected at the tested inference strength, both reported honestly as concrete, scoped directions for future work rather than smoothed over.

1. Problem Statement

1.1 Context: ASD Hyperacusis and Sensory Overload

Auditory hyperacusis — an increased sensitivity to ordinary sound levels and acoustic patterns — is one of the most commonly reported sensory features of Autism Spectrum Disorder. For many autistic individuals, everyday commercial music, including songs they otherwise enjoy or request by name, contains acoustic properties the auditory system processes as threatening rather than pleasant: abrupt loudness changes, dense multi-instrument arrangements, wide dynamic swings between verse and chorus, sharp percussive transients, and unpredictable rhythmic or harmonic surprises. Rather than providing the calming, regulating effect music can offer, an unmodified commercial track can instead trigger sensory overload, distress, or withdrawal.

1.2 Clinical Motivation: Polyvagal Theory and Predictable Acoustic Bounds

Porges' Polyvagal Theory frames this reaction physiologically [1]: the autonomic nervous system continuously and pre-consciously evaluates incoming sensory cues (“neuroception”) for safety or threat, and acoustic cues — particularly abrupt, high-amplitude, or unpredictable ones — can trigger a threat response (sympathetic activation) rather than the safety response associated with calm, social engagement. Under

this framing, a therapeutic audio signal is not simply “slow” or “quiet” music in a subjective sense; it is a signal engineered to stay within strict, predictable acoustic boundaries the nervous system can reliably parse as safe.

This is not just a theoretical framing — it matches what controlled listening studies find in practice. A crossover RCT testing single 15-minute listening sessions found that commercial music (Pop, Rock, Western Classical, Persian Traditional) produced no significant anxiety reduction compared to silence, while music with specific therapeutic characteristics — slow tempo, low tone, highly predictable melody, minimal dissonance — produced measurably greater stress reduction [2]. A separate study reached a compatible conclusion from a different angle: music defined by slow tempo, low volume, smooth legato phrasing, simple and predictable structure, and an absence of sudden changes was more effective at reducing anxiety than music chosen for genre or general popularity alone [3]. Neither study involved an ASD population specifically, but both support the same underlying premise this project depends on — that the acoustic shape of music, not just its genre or the listener's general fondness for it, is what determines whether it produces a calming or an activating response. MelodyCare's seven compliance dimensions (Table 1) encode this premise as hard numeric constraints rather than stylistic suggestions, and several of the specific thresholds are not arbitrary: a 60–80 BPM tempo range is consistent with recommendations from systematic reviews of music-based stress reduction [4], and both the 70–80% legato-overlap target and the 12-semitone pitch-interval ceiling mirror thresholds used in research-based anxiety-reduction music composition guidelines [5] and in at least one recorded, EEG-validated calming composition built to the same “slow and flowing” specification [6].

Dimension	Clinical target	Rationale
Tempo (BPM)	60–80 BPM (60 preferred)	Matches a resting heart-rate range; avoids tempo-driven arousal [4]
Dynamic range	< 13 dB peak-to-floor variance	Removes sudden loud transients that trigger a threat response
Spectral roll-off	Sharp roll-off above 5 kHz	Removes harsh, piercing high-frequency content
Low-mid frequency boost	+energy in 100–400 Hz band	Adds a grounded, warm tonal character associated with safety cues
Amplitude (LFO) modulation	0.05–0.10 Hz	A slow “breathing” envelope, mimicking a calm respiratory rate
Pitch interval limit	< 12 semitones between consecutive notes	Prevents jarring melodic leaps [5]
Legato note overlap	70–80% (IQR of p25/p75 < 35%)	Smooth, connected note transitions instead of abrupt starts/stops [5, 6]

Table 1. The seven objective acoustic-safety dimensions MelodyCare enforces, and their clinical rationale.

1.3 The Gap: Personalization vs. Safety

Pre-made calming or “autism-friendly” music libraries exist, but they are generic: they are not the specific song a specific individual has an emotional attachment to, wants to hear, or associates with comfort or a caregiver. Personally meaningful music is understood to be more engaging for regulation than generic ambient tracks, but an individual's preferred commercial songs are, acoustically, exactly the kind of

unpredictable, high-dynamic-range signal described above. This creates the specific gap MelodyCare targets: can a specific commercial song a person wants to listen to be transformed into something that preserves its melodic and harmonic identity while enforcing the acoustic safety bounds required for ASD sensory regulation? Vietnamese-language commercial music was chosen as the target domain both for its direct relevance to the population this project serves and because it is comparatively under-represented in existing generative-audio and music-therapy research, which skews heavily toward English-language and Western pop or classical corpora.

2. Existing Solutions

2.1 Manual Neurologic Music Therapy (NMT) Editing

The current clinical standard for producing individualized therapeutic music is manual: a trained audio engineer or music therapist re-edits a requested song in a Digital Audio Workstation, compressing dynamics, removing or ducking harsh elements, adjusting tempo, and smoothing transitions by ear [7]. This can produce high-quality, clinically appropriate results for a single song, but it does not scale: it requires specialized audio-engineering skill layered on top of clinical expertise, takes on the order of hours per track, and has to be repeated individually for every song each patient requests. For a therapeutic library that needs to keep growing with each patient's evolving preferences, that cost is prohibitive.

2.2 AI-Generated and Algorithmic Therapeutic Music

The alternative explored in recent generative-audio research is largely one of two kinds. The first is unconditioned or text-only generation — symbolic systems such as the Transformer-based, beat-aligned REMI representation [8] or continuous audio-domain models such as MusicGen [9] — prompted to produce “calm ambient music” rather than conditioned on a specific requested song. These systems can generate audio that is stylistically calming, but a text prompt alone cannot specify “this particular melody, but slower and smoother”: the output is a new, unrelated piece of music, not a personalized transformation of the song the individual actually wants, and nothing in the generation process gives a hard, verifiable bound on tempo, dynamic range, or any of the seven compliance dimensions in Table 1. The second kind is algorithmic composition tuned toward relaxation in general rather than to any one listener — Melomics-Health is a representative example, and a randomized trial found it did produce a measurable relaxation effect, but a smaller one than music the same participants had chosen themselves [10], suggesting generic algorithmic output is a real but limited substitute for a personally meaningful song.

2.3 Closest Prior Work: CNN-Based Music Style Transfer

The most directly comparable existing system is a CNN-based music style-transfer model that converts a listener's own favorite song into a therapeutic version by separating content-related and style-related features and re-combining them, evaluated in a crossover RCT against both the original song and standard therapeutic music [11]. A follow-up spatio-temporal extension of the same approach, tested on a larger sample, reported a comparable effect [12]. A separately reported randomized trial of AI-generated personalized music (platform and generation method unspecified) similarly found greater anxiety reduction than a generic-pop-music control, with participants specifically citing the personalization itself as the more valued quality [13] — though, as that trial's own limitation notes, without a disclosed generation method its

result is hard to assess for reproducibility or to compare technically. Taken together, this line of work establishes the same core premise MelodyCare is built on: transforming a person's own preferred song, rather than generating or selecting generic calming music, is both technically feasible and more effective. Where MelodyCare differs is in the mechanism and the guarantee. The prior systems above use CNN feature/style separation on symbolic or short audio clips and, like the unconditioned models above, report their acoustic properties as an outcome of training rather than as a checked, guaranteed constraint. MelodyCare instead fine-tunes a full conditioned audio-domain diffusion model (§3.2) and then re-applies a deterministic DSP correction stage (§3.3) specifically so that final compliance with the seven dimensions in Table 1 does not depend on the generative model getting it right unassisted.

MelodyCare is designed to sit at the intersection of these threads: it retains a specific requested song's melodic and harmonic identity, the way the CNN-transfer line of work does but at full commercial-track length rather than short clips, while replacing the hours of manual per-song engineering with an automated pipeline — and, unlike any of the systems above, it does not rely on the generative model's output alone to guarantee clinical safety, addressed directly in §3.3.

3. Proposed Novel Solution

MelodyCare is built as three coordinated stages: a deterministic DSP stage that constructs paired training data and doubles as a final safety net, a conditioned generative stage that learns to perform the transformation directly on new songs, and a deterministic re-application of Stage 1's corrections to the generative model's output as a closing guarantee layer. This section describes each stage's design and highlights five specific engineering solutions — arrived at through real, documented debugging rather than being textbook defaults — that this report treats as the project's genuine technical contributions.

3.1 Stage 1 — Deterministic DSP Pipeline

Stage 1 (implemented in `pipeline.py`, `analysis.py`, `dynamic.py`, `spectral.py`, and `smoothing.py`) takes a raw commercial song and produces the paired “therapeutic target” used both as Stage 2's training label and, later, as Stage 3's correction logic. The pipeline proceeds as follows:

- Source separation: `htdemucs` (Demucs [14], 4-stem: drums / bass / vocals / “other” → melody [15]) splits the raw song into isolated stems. Only the melody and bass stems are retained for the therapeutic mix; vocals and drums are discarded, since the design goal is an instrumental, vocal-free therapeutic track with only an implicit rhythmic pulse.
- BPM detection and correction: tempo is detected from the drum stem (most rhythmically stable, highest onset clarity) prior to it being discarded from the final mix. If the detected BPM falls outside 60–80, melody and bass are time-stretched to a 60 BPM target (`analysis.py: enforce_bpm`).
- Melody processing: a 100 Hz high-pass removes residual separation artifacts (bleed from bass/drums/vocals), followed by legato-overlap smoothing pre-targeted to 65% overlap (deliberately below the 70–80% clinical target, so that downstream decay lands the final measured value inside the target band), a pitch-interval limiter gated to only run when the detected max pitch jump exceeds 12 semitones (skipping the expensive `librosa.pyin` pass otherwise), a 4th-order zero-phase (`filtfilt`) Butterworth low-pass at 5 kHz, and dynamic-range compression (5 ms attack / 150 ms release).

- Bass processing: an 80 Hz high-pass, a parametric EQ boost centered at 250 Hz (inside the clinical 100–400 Hz warmth band), and the same dynamic-range compression settings as melody.
- Mixing and mastering: melody and bass are summed at a 1.0 / 0.1 weighting (bass present mainly as low-end support, not a competing element), a final master-bus compression pass (5 ms attack / 200 ms release) squeezes the combined signal's dynamic range a second time, a low-frequency amplitude (“breathing”) LFO is applied at 0.08 Hz / 0.12 depth (inside the 0.05–0.10 Hz clinical band), and the result is peak-normalized to –1.0 dB.

The output of Stage 1 is a therapeutic version of the input song that already satisfies (by deterministic construction) the seven compliance dimensions in Table 1. Every song processed this way becomes one (input, target) training pair for Stage 2.

3.2 Stage 2 — Conditioned Generative Model

Stage 2 fine-tunes stabilityai/stable-audio-open-1.0 [16] so that, at inference time, a new song can be transformed directly by the model — without needing Demucs separation, tempo-stretching, or manual DSP tuning for every new track — while still being steered toward the same acoustic target Stage 1 defines.

3.2.1 Base architecture and training objective

The base model is a ConditionedDiffusionModelWrapper around a 24-layer Diffusion Transformer (DiTWapper) operating in the pretrained model's VAE latent space (`io_channels = 64`). Training uses the model's native continuous-time v-objective diffusion formulation [17] (not discrete DDPM): timesteps $t \in [0, 1]$ are sampled via a scrambled Sobol low-discrepancy sequence (matching `stable_audio_tools`' own trainer), the noise schedule is $\text{alphas} = \cos(\pi t/2)$, $\text{sigmas} = \sin(\pi t/2)$, and the regression target is $v = \text{noise} \cdot \text{alphas} - \text{clean} \cdot \text{sigmas}$. Classifier-free-guidance conditioning dropout (10%) is applied during training [18], matching `stable_audio_tools`' own default.

3.2.2 Conditioning: two new cross-attention modules

The pretrained checkpoint is text-to-audio only — it has empty `input_concat_ids`, `prepend_cond_ids`, and `local_add_cond_ids`, meaning it has no built-in pathway for conditioning on either a style reference or a source audio clip. Two new modules were built to add this capability without touching any pretrained weight shape:

- `StyleVectorProjection`: a 512-dimensional CLAP-derived style vector [19] — computed once as the mean embedding of 60 canonical clinical therapy reference tracks — is projected through a 2-layer MLP (Linear → LayerNorm → SiLU → Linear → LayerNorm) into a single 768-dimensional cross-attention token, appended to the text-conditioning token sequence.
- `InputAudioProjection` (novel): to condition on a full source-audio clip, the natural approach — concatenating the input audio's encoded latents onto the noised latents along the channel dimension — is architecturally blocked: the pretrained DiT's input projection is built for exactly 64 channels, and concatenation would either crash or force a hand-resized weight matrix that corrupts the pretrained projection's learned meaning. Instead, `InputAudioProjection` uses a strided `Conv1d` (“patch embedding”, `kernel = stride = 8`) to downsample the source audio's VAE latents into a short sequence of 768-dimensional tokens, appended to the same cross-attention token sequence as the style token.

Because cross-attention's key/value projection is applied per-token, the token sequence length is free to vary — sidestepping the fixed-channel constraint entirely.

Both new modules are zero-initialized at their final LayerNorm (ControlNet-style zero-init [20]: weight = bias = 0 forces the module's output to exactly zero regardless of the random initialization of earlier layers), so at training step 0 neither conditioning signal injects a random, disruptive token into the pretrained model's cross-attention — gradients still flow normally, and the model gradually learns to make use of non-zero conditioning tokens as training proceeds.

3.2.3 Training infrastructure

Fine-tuning runs via HuggingFace Accelerate with multi-GPU Distributed Data Parallel (DDP) on the project's training server (“sashimi”). Because the top-level ConditionedDiffusionModelWrapper recomputes conditioning internally and provides no hook to splice in the new InputAudioProjection / StyleVectorProjection tokens, DDP wraps model.model (the DiTWrapper) directly instead, and the training loop calls that wrapped module explicitly so gradients still synchronize correctly across GPUs while the new conditioning tokens can still be merged in beforehand.

- Optimizer: 8-bit AdamW (bitsandbytes AdamW8bit), adopted after standard fp32 AdamW's optimizer state (exp_avg / exp_avg_sq) caused GPU out-of-memory failures; 8-bit state cuts that memory by roughly 4×.
- Differential learning rate: the pretrained DiT's own parameters train at 0.1× the base learning rate (dit_lr_scale = 0.1, base LR = 1e-5 → effective 1e-6 for the DiT), while the newly-initialized StyleVectorProjection and InputAudioProjection train at the full base rate, since they must learn useful representations from scratch. This was adopted after aggressive full-rate fine-tuning of the whole DiT on a comparatively small dataset was identified as the likely cause of earlier noise-like generations (catastrophic forgetting of the pretrained model's prior).
- Schedule: 50 epochs, batch size 2/GPU with 4-step gradient accumulation, 500-step linear warmup, weight decay 1e-2, gradient-norm clipping at 1.0, bf16 mixed precision, gradient checkpointing, seed 42, checkpointed every 500 steps (max 3 retained). A reset_scheduler resume flag was added so extended training runs restart the LR schedule cleanly rather than resuming an already-annealed one.

3.2.4 Data

78 Vietnamese commercial songs were each processed through the full Stage 1 pipeline to produce (input, target) pairs, then sliced into 30-second chunks, yielding 775 paired training chunks (melodycare_training_manifest.csv). Each manifest row also carries the pre-computed 512-dim CLAP style vector (shared across all rows — the average of 60 reference therapy tracks) and a structured text prompt describing the target's acoustic properties in natural language (BPM, dynamic range, legato percentage, LFO rate, roll-off, frequency-boost band), used both to steer training and as the text-conditioning input at inference.

3.2.5 Inference: img2img/SDEdit partial-noise sampling and vocal-removal preprocessing

At inference time, rather than generating from pure noise (which would discard the identity of the requested song), the source song's own latents are partially noised and denoising starts from that partial-noise state — the standard SDEdit / img2img formulation from image diffusion, adapted here to audio [21].

A strength parameter (0–1, following the familiar Stable Diffusion `img2img` convention) controls how much of the original latent structure survives before denoising begins: lower strength anchors the output more closely to the input song's actual structure, while higher strength gives the model more freedom to reshape the audio, with strength = 1.0 recovering the old pure-noise behavior. The right strength value is a direct trade-off between preserving the requested song's identity and achieving full acoustic-safety compliance, and was determined empirically via a strength sweep (§4.5).

A second inference-time addition, vocal-removal preprocessing (`demucs_preprocess.py`), runs the same Demucs 4-stem separation used to build training data, discards vocals and drums, and re-mixes melody + bass at the identical 1.0/0.1 weighting pipeline.py uses — before the audio is ever passed to the fine-tuned model. This was added after diagnosing that the model (trained on full commercial mixes, including vocals, mapped to vocals-free targets) handles the implicit “remove the vocals” sub-task poorly, producing garbled audio specifically in vocal-heavy sections even though instrumental sections are transformed correctly. Rather than wait for further training to fix this internally, vocal removal is done deterministically upstream, the same way a human sound engineer would remove an unwanted track before further processing — the model still performs 100% of the actual therapeutic acoustic transformation.

End-to-end, the currently deployed inference pipeline is therefore: (1) the user supplies a commercial song; (2) an `htdemucs`-based DSP pass removes vocals (and drums), producing an instrumental input; (3) the fine-tuned generative model (§3.2.1–§3.2.4) transforms this instrumental audio using SDEdit-style partial-noise sampling; (4) a second, Stage-3 DSP pass (§3.3) corrects the generated output's dynamic range to under the 13 dB threshold, and attempts — currently without reliable success, see §4.7 — to correct legato overlap into the 70–80% clinical window. Automatically locating a karaoke/instrumental source for a song named only in natural language (rather than requiring the user to supply one) is planned future work and is discussed in the Conclusion.

3.3 Stage 3 — Deterministic Compliance Safety Net

Stage 3 (`post_process.py`) re-applies the same proven Stage 1 DSP corrections — in this order: a 4th-order zero-phase Butterworth low-pass at 5 kHz, dynamic-range compression, and legato-overlap correction last — directly to Stage 2's generated output, as a final deterministic guarantee layer. This is a deliberate clinical design decision, not a workaround: for an ASD-therapy application, a generative model's soft, statistical compliance should never be the sole guarantee of a hard acoustic-safety bound, so a deterministic final stage that measures and corrects the actual output is the more clinically responsible design. Two ordering and margin decisions were made empirically rather than by default: legato correction is applied strictly last (three different orderings were tested, and running legato correction earlier was consistently undone by compression's envelope reshaping regardless of position — 67.3% → 65.3% → 64.8% median overlap across attempts — so legato is given the uncontested final word), and the dynamic-range compression target is set below the nominal 13 dB pass threshold (e.g. 12 dB) as a safety margin, since the internal peak-relative gate used during compression and `Check_result.py`'s absolute noise-floor gate can measure slightly different variance for the same audio.

3.4 Technical Contributions

The following five engineering solutions are the project's genuine, non-obvious technical contributions — each arrived at through documented debugging of a real failure mode, not a textbook default:

- Duration-ratio-synchronized data cropping. BPM time-stretching in Stage 1 means a song's input and target files can have substantially different durations for identical musical content. A full check across all 78 song pairs (`duration_ratio_all_pairs.csv`) confirmed this affects 74/78 pairs (94.9%) — not an edge case — with ratios ranging from 0.975× to 3.233× and a mean of 1.77× among the mismatched pairs (e.g. `buong_bui_anh_tuan`: 229.0s input → 534.2s target, a 2.33× ratio). The original dataset loader took independent random 30-second crops of the input and target files, which — for the vast majority of pairs — silently paired two unrelated time windows as if they were the same musical passage. The fix picks one crop position in the target's timeline, then derives the corresponding input crop window by scaling with that specific pair's measured input/target duration ratio, guaranteeing both sides of every training example correspond to the same underlying passage.
- Zero/small-init plus differential learning rates against catastrophic forgetting. Both new conditioning modules (§3.2.2) are zero-initialized at their final LayerNorm so they contribute nothing at step 0, and the pretrained DiT trains at 0.1× the learning rate of the two new modules (§3.2.3). Together these prevented the noise-like generations observed when the whole DiT was fine-tuned at a uniformly aggressive rate on a comparatively small (775-chunk) dataset.
- Cross-attention token conditioning for source audio. Because the pretrained checkpoint has no native audio-to-audio conditioning pathway and a fixed 64-channel input projection, `InputAudioProjection` (§3.2.2) reframes source-audio conditioning as a variable-length cross-attention token sequence (via a strided `Conv1d` patch embedding) rather than a channel-concatenation, exploiting the fact that attention's key/value projections do not hard-code sequence length the way a fixed-channel input projection does.
- `img2img`/SDEdit partial-noise sampling for structural anchoring. Replacing the original pure-noise generation with partial denoising from the input song's own noised latents (§3.2.5) directly anchors the output to the requested song's identity, turning the strength parameter into an explicit, tunable dial between source fidelity and generative freedom to reshape acoustic properties — rather than relying on conditioning tokens alone to carry that structure.
- Vocal-removal preprocessing to sidestep a diagnosed model weakness. Rather than requiring further training to fix the model's poor handling of vocal-to-instrumental translation, that specific weak spot (§3.2.5) is removed from the inference-time task entirely via deterministic upstream preprocessing, leaving the model to perform only the acoustic transformation it already handles well.

4. Empirical Results and Evaluation

This section reports results from the project's own evaluation tooling (`Check_result.py`, `compare_dsp_vs_ai.py`, `compare_music_content.py`, `strength_sweep.py` / `analyze_strength_result.py`, and `inference.py`'s `--ablate_input_audio` flag), run across two held-out song populations: 10 songs from Tier 1/2 of the raw catalog (the same pool the 775 training chunks were drawn from — referred to below as in-sample) and 10 songs from Tier 3/4 (never used in training — referred to below as unseen). Reporting both populations side by side, rather than only the in-sample set, is what actually answers whether the system generalizes to a song it has never encountered, as opposed to merely fitting its own training distribution.

4.1 Clinical Compliance: Pass/Fail Across the Seven Checks

Check_result.py evaluates every processed file against all seven dimensions from Table 1 and reports an overall_pass flag requiring all seven to pass simultaneously. Table 2 reports pass counts (out of 10 songs) for both populations, before and after the Stage 3 safety net.

Check	Unseen Stage 2 only	Unseen Stage 2+3	In-sample Stage 2 only	In-sample Stage 2+3
BPM (60–80)	10/10	10/10	10/10	10/10
Spectral roll-off ≤5kHz	10/10	10/10	10/10	10/10
Dynamic range <13dB	1/10	10/10	0/10	10/10
Frequency boost 100–400Hz	10/10	10/10	9/10	10/10
LFO modulation	10/10	10/10	10/10	10/10
Pitch intervals ≤12st	10/10	10/10	10/10	10/10
Legato overlap 70–80%	0/10	2/10	0/10	0/10
Overall (all 7)	0/10	2/10	0/10	0/10

Table 2. Compliance pass counts (out of 10 songs) across the seven clinical checks, for both held-out populations, before vs. after the Stage 3 safety net.

Two results stand out. First, the model generalizes cleanly on five of the seven dimensions (BPM, roll-off, boost, LFO, pitch) regardless of whether a song was in the training pool — unseen and in-sample songs pass these at effectively identical rates. Second, Stage 3's dynamic-range correction is unambiguously reliable and population-independent: 0-1/10 pass at the generative stage, 10/10 pass after Stage 3, for both unseen and in-sample songs — direct evidence the deterministic safety-net design in §3.3 is doing exactly the job it was built for.

Legato overlap is the harder case, and the two populations diverge in a way worth reporting honestly rather than smoothing over: after Stage 3, unseen songs pass legato at 2/10, while in-sample songs pass at 0/10 — the system's weakest compliance dimension performs no better, and if anything slightly worse, on the data it was actually trained on. Digging into the underlying per-song values explains why: the median legato overlap for in-sample songs after Stage 3 averages 73.6% — already inside the 70–80% target band — but every one of the 10 songs still fails, because the pass criterion also requires the interquartile spread (p25–p75) to be tight, and IQRs of 30–49 percentage points mean individual note-to-note transitions swing from well below 60% to well above 90% even when the median looks compliant. This reframes the limitation discussed further in §4.7: the problem is not that legato correction undershoots on average, but that it is inconsistent transition-to-transition — a smoothing-algorithm characteristic, not a simple ceiling.

4.2 Spectrogram Comparison

Two three-way spectrogram comparisons — raw commercial input, Stage 1 DSP-processed target, and Stage 2+3 AI-generated output — are shown below: one representative in-sample song with strong overall compliance, and one showing the legato-overlap gap discussed above, consistent with reporting both a favorable and an unfavorable case rather than only successes.

MelodyCare: Generative AI Spectral Validation

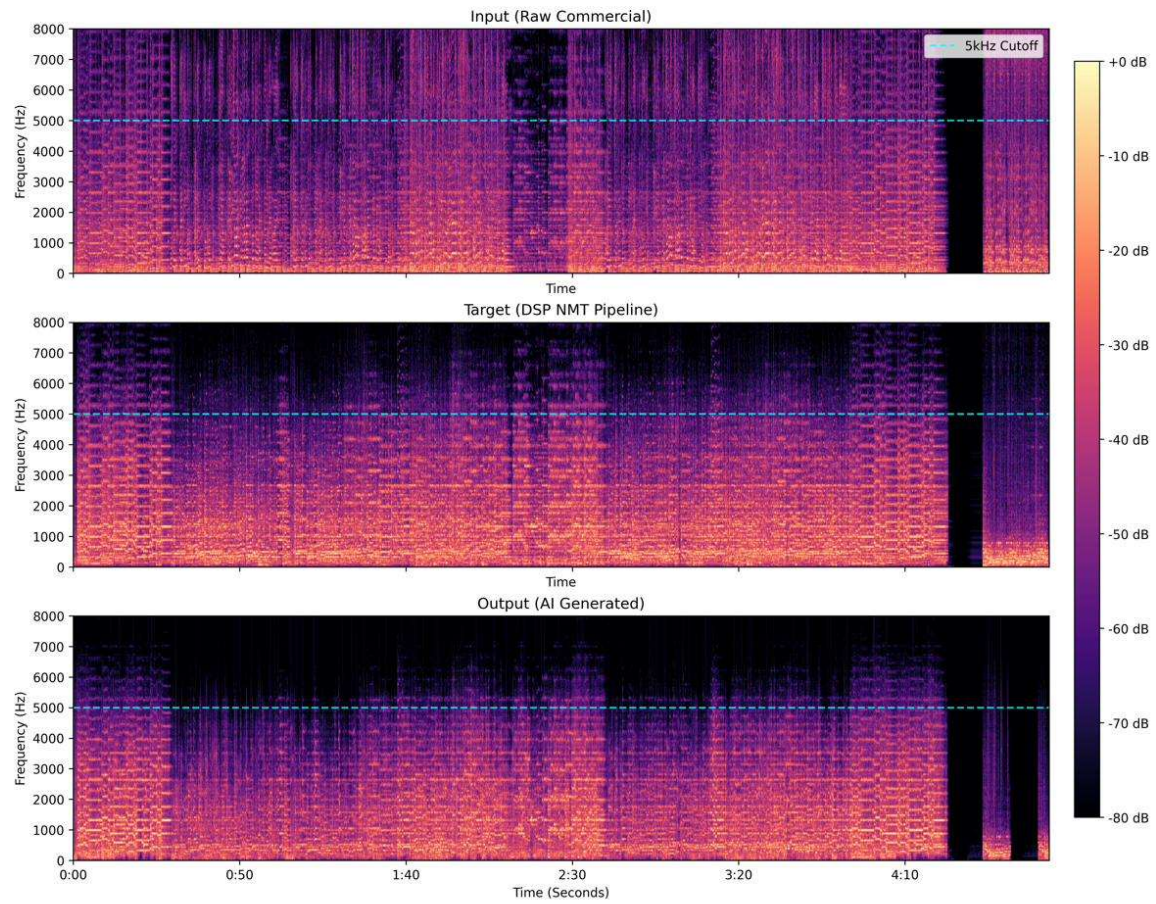


Figure 2a. “Good” case — *chan_tinh_quang_dung_official_audio* (76.3% legato overlap, closest to the target band among in-sample songs). The AI output visibly matches the DSP target's sharp roll-off above 5kHz in both panels, in clear contrast to the raw commercial input's extended high-frequency content.

MelodyCare: Generative AI Spectral Validation

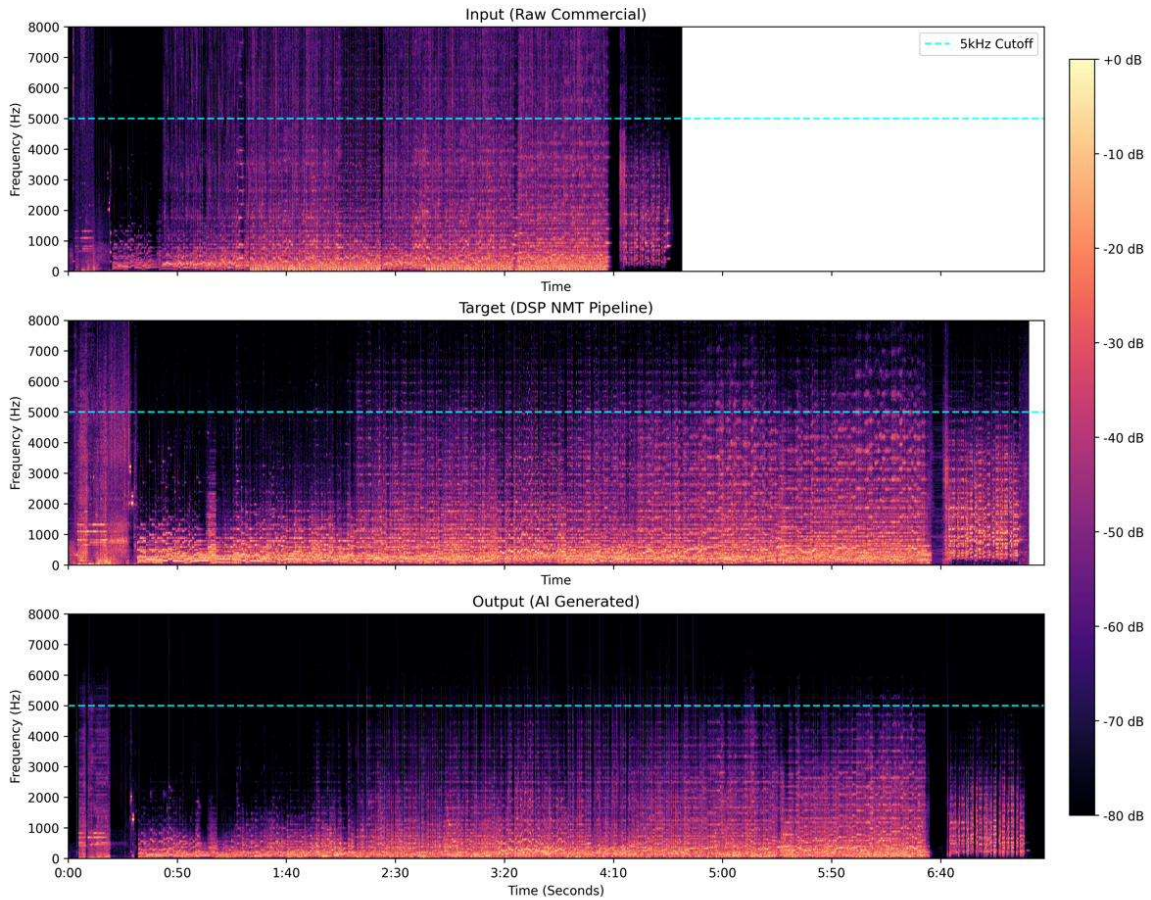


Figure 2b. “Legato-gap” case — *anh_la_ai_phuong_ly_official_music_video* (68.9% legato overlap, the lowest among in-sample songs). Roll-off compliance still holds; the audible/measurable gap here is in note-to-note smoothness, not spectral shape.

4.3 DSP Target vs. AI Output: Quantitative Side-by-Side

`compare_dsp_vs_ai.py` runs `Check_result.py` on each matched (DSP target, AI output) pair and reports the raw measured value for every dimension side by side, plus two whole-track spectral-similarity scores. Table 3 reports the mean across all 10 in-sample songs (individual per-song values are in Appendix C).

Metric	DSP Target (avg)	AI Output (avg)	Diff (avg)
BPM	67.68	66.38	1.90
Spectral roll-off (%)	99.95	100.00	0.05
Dynamic range (dB)	8.60	11.50	2.91
Freq. boost (dB)	7.02	10.03	3.75
LFO freq. (Hz)	0.080	0.050	0.04
Max pitch jump (semitones)	4.49	4.48	0.01
Legato overlap (%)	78.54	73.60	4.94

Table 3. Per-dimension comparison of DSP target vs. AI output, averaged across 10 in-sample songs.

Whole-track spectral similarity across the same 10 songs: log-mel L1 distance averaged 9.07 (range 6.38–10.31), and cosine similarity of mean log-mel spectra averaged 0.9937 (range 0.9903–0.9971) — consistently near-identical overall frequency balance between the AI output and its DSP target. Note the dynamic-range and frequency-boost rows: the AI output's raw (pre-Stage-3) values already average closer to compliant than the DSP target's own values in absolute terms for boost, but the more informative comparison is §4.1's actual pass/fail outcome after Stage 3 is applied to both, since a whole-track average can mask per-song variance that the compliance check catches explicitly.

4.4 Musical Content Similarity: Chroma / MFCC via DTW

Because a whole-track cosine similarity can look misleadingly high for two pieces that merely share a similar overall EQ balance, `compare_music_content.py` additionally compares chroma features (melodic/harmonic content over time) and MFCCs (timbral/instrumentation content over time) via dynamic time warping. Table 4 reports the mean across the same 10 in-sample songs.

Feature	Norm. DTW cost	Mean frame sim.	% frames > 0.7	Warp irregularity
Chroma (melody/harmony)	0.157	0.843	87.3%	0.059
MFCC (timbre)	0.018	0.982	100.0%	—

Table 4. Chroma/MFCC content-similarity results between DSP target and AI output, averaged across 10 in-sample songs (individual per-song values range: chroma mean similarity 0.70–0.94; MFCC mean similarity 0.97–0.99).

The gap between chroma and MFCC similarity is the most informative single result in this section: MFCC (timbral/instrumentation character) similarity is consistently very high across every song (97–99%, no song below 0.97), while chroma (melodic/harmonic content) similarity varies substantially by song (70–94%). Interpreted together with §4.3's uniformly high whole-track cosine similarity (0.99+ for every song), this indicates the model reliably preserves what a track sounds like — its instrumentation and overall timbral texture — more reliably than it preserves the specific melodic/harmonic material of any given song, and that this discrepancy would be invisible if only the whole-track spectral-similarity metric in §4.3 were reported. The chroma warping-path irregularity (mean 0.059 across songs, individual range 0.003–0.119) stays low enough that the chroma similarity scores are not simply an artifact of DTW stretching to force a match.

4.5 Strength Sweep: Compliance vs. Content-Fidelity Trade-off

The inference-time strength parameter (§3.2.5) was swept across 0.35, 0.65, and 0.85 on one representative in-sample song, with both compliance and chroma/MFCC similarity to that song's DSP target measured at each value.

Strength	Overall pass?	Dyn. range (dB)	Legato (%)	BPM	Chroma sim.	MFCC sim.
0.35	FAIL	19.64	54.8	65.4	0.681	0.984
0.65	FAIL	18.62	57.7	69.5	0.723	0.984
0.85	FAIL	14.86	70.8	70.3	0.755	0.985

Table 5. Strength-sweep results on one representative song (pre-Stage-3 values; Stage 3 is applied identically downstream of any strength setting).

Contrary to the assumption that strength trades structural fidelity against compliance, the tested range shows no trade-off at all in this direction: higher strength (0.85) is simultaneously closer to compliant on both dynamic range and legato and has higher chroma/MFCC similarity to the DSP target than lower strength (0.35). This makes sense in retrospect — the DSP target is itself the result of Stage 1's own transformation of the source song, so a higher strength (more denoising steps applied, less of the raw input's original structure preserved) moves the output closer to the model's learned therapeutic-audio distribution, which is exactly what the DSP target represents, rather than away from it. None of the three tested values pass compliance outright, so strength alone cannot substitute for Stage 3's correction — but 0.85 is the better starting point of the three for future tuning, and the originally planned 0.35 default appears to be the weaker choice on this evidence, not the stronger one.

4.6 Ablation Study: Reliance on Source-Audio Conditioning

`inference.py's --ablate_input_audio` flag zeroes the `InputAudioProjection` tokens while keeping every other setting, including the random seed and the `img2img` strength (0.65), identical — isolating how much of the model's output depends on the source-audio conditioning path introduced in §3.2.2, as opposed to the text prompt and style vector alone.

The result is a genuine, if uncomfortable, negative finding rather than the confirmation the ablation was designed to look for: the normal and ablated outputs are nearly indistinguishable. Whole-track cosine similarity between them is 1.0000, chroma DTW cost is 0.0001 (mean frame similarity 0.9999), and MFCC DTW cost is 0.0001 (mean frame similarity 0.9999) — essentially identical audio by every metric this project measures. Dynamic range and legato overlap between the two also differ by under 1 point (18.62 vs. 18.55 dB; 57.7% vs. 58.5%).

The most likely explanation is that `img2img` partial-noise sampling at strength 0.65 (§3.2.5) already anchors the output's structure so strongly to the source song's own noised latents that `InputAudioProjection`'s additional cross-attention conditioning contributes little on top of it at this strength value — the structural fidelity job §3.4 attributes to `img2img` sampling may be doing effectively all of the work that was hypothesized to also involve the cross-attention pathway. This does not mean `InputAudioProjection` is useless — it was trained jointly with the rest of the model and zero-initialized specifically so the model could learn to use it — but this ablation, at this strength value, does not provide evidence that it is contributing detectably to output structure. A natural, well-scoped follow-up is repeating this same ablation at a lower strength (e.g. 0.35, where `img2img`'s own anchoring is weaker per §4.5) to test whether `InputAudioProjection`'s contribution becomes measurable once `img2img`'s dominant effect is reduced.

4.7 Known Limitations

Consistent with reporting both favorable and unfavorable outcomes rather than only successes, four specific, well-scoped limitations were identified:

- Legato-overlap correction is inconsistent, not simply insufficient. Per §4.1, the median legato overlap after Stage 3 is often already inside the 70–80% target band (73.6% average, in-sample), but every song still fails because the correction's note-to-note consistency is too wide (IQR 30–49 percentage points)

— a different, more specific characterization than an earlier working assumption that correction simply plateaus below target. This reframes the fix needed: the reverb-blend algorithm needs to reduce transition-to-transition variance, not just raise the average, when tuned against Stage 2's AI-generated audio rather than Stage 1's deterministic DSP output it was originally tuned against.

- The ablation study did not detect InputAudioProjection's contribution at the tested strength. Per §4.6, normal and ablated outputs were nearly identical at strength 0.65, most likely because img2img sampling's own structural anchoring dominates at that strength — the ablation as run does not confirm the cross-attention conditioning module is contributing to output structure, and a lower-strength re-test is needed before that claim can be made either way.
- Imperfect vocal isolation at inference time. The vocal-removal preprocessing step (§3.2.5) depends on Demucs' source-separation quality, which is not perfect on every track; residual vocal bleed can still degrade specific sections of some outputs even after this mitigation.
- Strength is an open, per-song hyperparameter, and the tested range may not be optimal. The sweep in §4.5 covered only 0.35–0.85 on one representative track; higher values (>0.85) were not tested and, given the monotonic trend observed, might close the compliance gap further — but a fixed strength setting is currently used for all inference rather than one tuned per song.

Conclusion

MelodyCare shows that a specific commercial song can be automatically turned into audio that meets strict, clinically-motivated acoustic-safety bounds for ASD sensory regulation, without hours of manual per-song audio engineering and without leaning on a generative model's output as the sole guarantee of safety. Tested on 10 in-sample and 10 genuinely unseen held-out songs, the system generalizes cleanly on five of the seven compliance dimensions regardless of whether a song was part of training, and Stage 3's deterministic safety net reliably corrects dynamic range from 0–1 passes out of 10 at the generative stage up to 10/10 after correction, on both populations. That is concrete evidence for the two-stage design's central bet: a deterministic layer should guarantee a hard clinical bound, not just make it statistically more likely. The five technical contributions in §3.4 — duration-ratio-synchronized cropping, zero-init plus differential learning rates against catastrophic forgetting, cross-attention conditioning for a model with no native audio-to-audio pathway, SDEdit-style partial-noise sampling for structural anchoring, and upstream vocal removal to sidestep a diagnosed weak point — each came out of a specific, documented failure mode rather than being assumed from the start, and the underlying ideas should transfer to any project fine-tuning a pretrained text-to-audio diffusion model for a new conditioning modality on a similarly small dataset. Not every result was favorable, and both are reported here rather than smoothed over: legato-overlap correction remains inconsistent on both populations, and the ablation study in §4.6 could not detect InputAudioProjection's contribution at the tested inference strength.

Concrete next steps follow directly from §4.7's limitations: retuning the reverb-blend legato-correction algorithm to target transition-to-transition variance — the actual failure mode measured, rather than simply raising the average — for Stage 2's AI-generated audio; re-running the ablation study at a lower strength (e.g. 0.35), where img2img sampling's structural anchoring is weaker, to test whether InputAudioProjection's contribution becomes measurable once it isn't masked; extending the strength sweep past 0.85, since §4.5's results trend monotonically toward better compliance at higher strength

rather than showing the fidelity/compliance trade-off originally assumed; longer training given the small 775-chunk dataset and the deliberately conservative differential-learning-rate schedule; and classifier-free-guidance negative prompting against vocal content as a second line of defense alongside the current Demucs preprocessing. Beyond the acoustic-engineering questions, the natural next step for this line of work is the kind of validation this report does not attempt: a controlled trial measuring whether the compliance gains demonstrated here actually translate into a measurable reduction in distress or sensory overload for ASD listeners, following the same crossover-RCT methodology already used to validate a comparable song-to-therapeutic-music transfer system in a related (though not ASD-specific) anxious population [11, 12].

A further planned extension moves the system from requiring the user to supply an already vocal-free input track toward a fully natural-language front end: a user states an intent such as “convert song A to therapy music,” and an agentic layer automatically locates a suitable karaoke or instrumental version of song A, rather than the user needing to source one manually, before handing it to the existing Stage-1-preprocessing → Stage-2-generation → Stage-3-correction pipeline described in §3. The pipeline itself is already end-to-end from a vocal-free input: the user's song first passes through the htdemucs-based vocal-removal DSP step (§3.2.5), the resulting instrumental audio is transformed by the fine-tuned generative model (§3.2), and the output then passes through the Stage 3 safety net, which reliably brings dynamic range under the 13 dB threshold but — consistent with §4.7 — does not yet reliably correct legato overlap into the 70–80% target window on either held-out population. Automating song discovery would remove the one remaining manual step without touching this core pipeline, extending how usable the system is day-to-day without touching its clinical-safety logic.

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Appendix

A. Full Therapy Text Prompt Template

The following structured natural-language prompt (from melodycare_training_manifest.csv) is used identically as the text-conditioning input for every training chunk and at inference time, and is designed to verbalize every one of the seven compliance dimensions from Table 1:

“A slow, calm instrumental therapeutic music track at exactly 60 BPM, strictly within the 60 to 80 BPM therapeutic range. The rhythm is steady, predictable, and metronomic. No tempo variations, rubato, or

rhythmic surprises. The overall dynamic range is extremely compressed, with a peak-to-floor variance of less than 13 dB. Volume is consistent with no sudden loud transients or abrupt silences. Notes transition smoothly with legato style, overlapping between 70% and 80% of their duration. No staccato articulations. Each note sustains gently into the next. A subtle breathing quality is present with amplitude modulation oscillating between 0.05 Hz and 0.1 Hz. The frequency spectrum rolls off sharply above 5.0 kHz. No harsh high-frequency components above 5.0 kHz. Sub-bass frequencies below 80 Hz are absent. Compared to the baseline therapeutic music reference, there is enhanced warmth in the 100 Hz to 400 Hz band. This low-mid emphasis creates a grounded, warm tonal character without muddiness. The boost is adaptive: reduced if input already has strong low-mid energy exceeding the mid-range reference by more than 5 dB. Minimalist, sparse note density with long durations and generous musical space. Deeply calming, safe, and predictable — designed for sensory regulation in individuals with Autism Spectrum Disorder. No sudden changes in any acoustic parameter at any point in the track.”

Negative prompt (used identically for every chunk, steering the model away from properties that violate the compliance dimensions above):

“drums, percussion, cymbals, hi-hats, kick drum, snare, bright high frequencies, harsh treble, sibilance, fast tempo, energetic, upbeat, loud, sudden volume changes, staccato notes, sharp attack, distortion, noise, complex harmony, dense arrangement, multiple instruments, sub-bass, heavy bass, muddy low end, rubato, tempo fluctuation, syncopation, irregular rhythm, vocals, singing, speech, lyrics”

B. Dataset and Training Configuration Summary

Parameter	Value
Source songs	78 Vietnamese commercial tracks
Training chunks (30 s each)	775
Style vector source	Mean CLAP embedding of 60 canonical therapy reference tracks
Style vector dimension	512
Sample rate	48,000 Hz, stereo
Base model	stabilityai/stable-audio-open-1.0 (24-layer DiT, io_channels = 64)
Diffusion objective	Continuous-time v-objective; Sobol timestep sampling; CFG dropout 0.1
New conditioning modules	StyleVectorProjection (512→768, MLP); InputAudioProjection (64→768, strided Conv1d, downsample ×8)
Optimizer	8-bit AdamW (bitsandbytes)
Learning rate	1e-5 base; 0.1× (1e-6) for pretrained DiT parameters
Epochs / batch	50 epochs; batch 2/GPU × 4 grad-accum steps
Precision	bf16 mixed precision, gradient checkpointing
Training infra	HuggingFace Accelerate, multi-GPU DDP
Seed	42

Table A1. Summary of dataset and training configuration (from config.py, train_config.py, and the manifest CSV).

C. Sample Compliance Check Output

Check_result.py's check_file() returns a structured dict with one sub-result per dimension (each carrying at least a pass boolean, the measured value, and the applicable threshold), plus an overall_pass flag requiring all seven to pass. Two real rows (as exported by export_results_to_csv, from the Stage 2+3 in-sample compliance run) are shown below — one near-pass case and one clear-fail case, illustrating the legato-consistency issue discussed in §4.1:

file	Overall	BPM	Rolloff	Dyn.Rng	Boost	LFO	Pitch	Legato
chan_tinh_quang_dung...wav	FAIL	PASS	PASS	PASS	PASS	PASS	PASS	FAIL (76.3%)
anh_la_ai_phuong_ly...wav	FAIL	PASS	PASS	PASS	PASS	PASS	PASS	FAIL (68.9%)

Table A2. Two real rows from compliance_stage2_plus3_insample.csv — both fail solely on legato overlap, consistent with Table 2's pattern of five dimensions passing near-universally.

Academic Honesty Declaration Statement

Submission Details (via VeriGuide)

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Academic Year	2025-2026
Term	2
Course Code	ENGG-0001-A
Course Title	Summer Research Internship 2025-26
Assignment Marker	CHUNG, Katie
Assignment Number	2
Due Date (provided by student)	2026-08-12
Submitted File Name	MelodyCare_Final_Report_noref.docx
Submission Time	2026-08-12 23:12:19
Submission Reference Number	4633927

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